



Seminar 2

2.2. Decide if the given points are collinear;

1) $P(3, -5), Q(-1, 2), R(-5, 5)$

2) $P(1, 0, -1), Q(0, -1, 2), R(-1, 2, 5)$

1) P, Q, R - collinear $\Rightarrow$! Only 2D!

$$\Rightarrow \begin{vmatrix} x_P & y_P & 1 \\ x_Q & y_Q & 1 \\ x_R & y_R & 1 \end{vmatrix} = 0 \Leftrightarrow \begin{vmatrix} 3 & -5 & 1 \\ -1 & 2 & 1 \\ -5 & 5 & 1 \end{vmatrix} = 0$$

$$\Leftrightarrow 6 - 5 + 25 - (-10 + 25 + 5) = 0$$

$$\Leftrightarrow 22 - 22 = 0$$

$$0 = 0 \text{ TRUE } \Rightarrow$$

$\Rightarrow P, Q, R$ - collinear

2) $\vec{PQ} = \vec{r}_Q - \vec{r}_P$

$$= (0, -1, 2) - (1, 0, -1) = (-1, -1, 1)$$

$$\vec{PR} = \vec{r}_R - \vec{r}_P = (-1, 2, 5) - (1, 0, -1)$$

$$= (-2, -2, 6) =$$

$$= 2 \cdot (-1, -1, 3) = 2 \cdot \vec{PQ} \Rightarrow$$

$$\Rightarrow P-Q-R \text{ collinear}$$

why the 1st approach works

$$\left| \begin{array}{ccc|c} x_P & y_P & 1 & L_2 \leftarrow L_2 - L_1 \\ x_Q & y_Q & 1 & = \\ x_R & y_R & 1 & L_3 \leftarrow L_3 - L_1 \end{array} \right|$$

$$= \left| \begin{array}{cccc} x_P & y_P & 1 & \\ x_Q - x_P & y_Q - y_P & 0 & \\ x_R - x_P & y_R - y_P & 0 & \end{array} \right|$$

$$= \left| \begin{array}{cc} \vec{x}_{PQ} & \vec{y}_{PQ} \\ \vec{x}_{PR} & \vec{y}_{PR} \end{array} \right| = 0 \Leftrightarrow \vec{PR} = 2\vec{PQ}$$

Changing reference frames (Euclidean space)
 $\hookrightarrow$ "space where we do geometry"

V - associated v.v.

$$V = \frac{\|E\| \times \|E\|}{\sim}$$

$\leadsto$ Basically says:
 "Taking
 is the same as taking"

V - k -v.s.

$$\left. \begin{aligned} b &= (v_1, \dots, v_n) \\ b' &= (v'_1, \dots, v'_n) \end{aligned} \right\} \text{bases of } V$$

$$\begin{aligned} M_{b'b} &= [id]_{b'b} = \left([v_1]_{b'} \mid \dots \mid [v_n]_{b'} \right) \\ &= \text{Base change matrix } b \rightarrow b' \end{aligned}$$

$\forall v \in V:$

$$[v]_{b'} = M_{b'b} \cdot [v]_b$$

Reference frame = a pair (O, b) where:

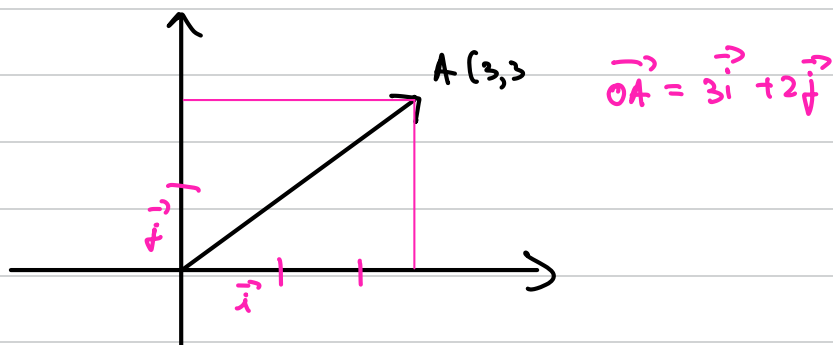
- $O \in \mathbb{E}$ - point
- b - base of V

$K = (O, b)$ - reference frame

$\forall P \in \mathbb{E}$

$$[P]_K := [\vec{OP}]_b$$

Ex:



$$K = (O, b)$$

$$K' = (O', b')$$

$$\begin{aligned} [P]_{K'} &= [\overrightarrow{O'P}]_{b'} = M_{b'b} [\overrightarrow{O'P}]_b \\ &= M_{b'b} \cdot [\overrightarrow{OP} - \overrightarrow{OO'}]_b = \\ &= M_{b'b} \cdot ([\overrightarrow{OP}]_b - [\overrightarrow{OO'}]_b) \\ &= M_{b'b} \cdot ([P]_K - [O']_K) \end{aligned}$$



$M_{K'K}$ (we will use it more)

$$2.6. \left. \begin{array}{l} K = (O, B) \\ K' = (O', B') \end{array} \right\} \text{Reference frames in } \mathbb{E}^2 \rightarrow \text{given dim., NOT } \mathbb{E} \times \mathbb{E}$$

$$B = (\vec{i}, \vec{j})$$

$$B' = (\vec{i}', \vec{j}')$$

$$[O']_K = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

$$\{i'\}_B = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\{i'\}_B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$M_{K'K} = ?$$

coordinates of A, B, C in E^2 :

$$\{A\}_K = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$\{B\}_K = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

$$\{C\}_K = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$M_{KK'} = M_{BB'} = \begin{pmatrix} -3 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|cc} -3 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ -3 & 1 & 1 & 0 \end{array} \right) \quad L_1 \leftrightarrow L_2$$

$$\sim \begin{array}{l} L_2 \leftarrow L_2 + 3L_1 \end{array} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 4 & 1 & 3 \end{array} \right)$$

$$\sim \begin{array}{l} L_2 \leftarrow \frac{L_2}{4} \end{array} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & \frac{1}{4} & \frac{3}{4} \end{array} \right)$$

$$\sim \begin{array}{l} L_1 \leftarrow L_1 - L_2 \end{array} \left(\begin{array}{cc|cc} 1 & 0 & -\frac{1}{4} & \frac{1}{4} \\ 0 & 1 & \frac{1}{4} & \frac{3}{4} \end{array} \right)$$

$$\Rightarrow (M_{B'B})^{-1} = M_{B'B} = M_{K'K} = \begin{pmatrix} -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

$$\begin{aligned} \text{Let } P \in \mathbb{E}^2, \quad [P]_{K'} &= [\vec{OP}]_{B'} = M_{B'B} \cdot [O'P]_B = \\ &= M_{B'B} \cdot ([\vec{OP}]_B - [\vec{OO'}]_B) \\ &= M_{B'B} \cdot ([\vec{OP}]_B - [00']_B) \\ &= M_{K'K} \cdot ([P]_K - [O']_K) = \\ &\quad \hookrightarrow \begin{pmatrix} 6 \\ 0 \end{pmatrix} \end{aligned}$$

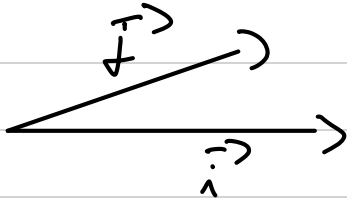
We can do:

$$[ABC]_{K'} = M_{K'K} ([ABC]_K - [O'O'O']_{K'})$$

The same thing as doing it 3 times

$$\begin{aligned} [ABC]_{K'} &= \begin{bmatrix} -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} \cdot \left(\begin{bmatrix} 0 & 6 & 3 \\ 2 & 4 & 1 \end{bmatrix} - \begin{bmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} \cdot \begin{bmatrix} -6 & 0 & -3 \\ 2 & 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 1 & 1 \\ 0 & 3 & 0 \end{bmatrix} \Rightarrow \begin{aligned} [A]_{K'} &= \begin{bmatrix} -1 \\ 0 \end{bmatrix} \\ [B]_{K'} &= \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ [C]_{K'} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{aligned}$$

$\mathbb{R}^2$



$(0, \vec{i}, \vec{j})$ - right oriented if we can rotate $\vec{i}$ by a positive angle in $[0, 2\pi)$ to overlap it with $\vec{j}$

left oriented - $\dots (-2\pi, 0) \dots$

2 ref. frames $\mathcal{K}, \mathcal{K}'$ have the same orientation i.f.f.
 $\det M_{\mathcal{K}\mathcal{K}'} > 0$

2.5. With respect to the basis $\mathcal{B} = (\vec{i}, \vec{j}, \vec{k})$:

$$\begin{aligned}\vec{u} &= \vec{i} + \vec{j} \\ \vec{v} &= \vec{j} + \vec{k} \\ \vec{w} &= \vec{i} + \vec{k}\end{aligned}$$

check that $\mathcal{B}' = (\vec{u}, \vec{v}, \vec{w})$ is a basis and decide if it's left / right oriented

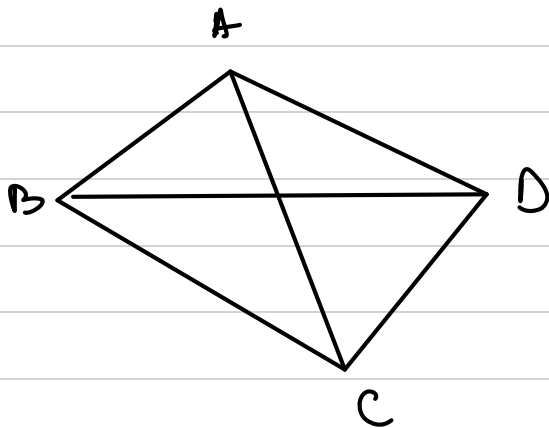
$$\begin{aligned}\vec{u} &= \vec{i} + \vec{j} \Rightarrow [\vec{u}]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ \vec{v} &= \vec{j} + \vec{k} \Rightarrow [\vec{v}]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\ \vec{w} &= \vec{i} + \vec{k} \Rightarrow [\vec{w}]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\end{aligned}$$

$$M_{B'B} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\det(M_{B'B}) = 1 + 1 = 2 \neq 0 \Rightarrow B' \text{ - basis}$$

$$\det(M_{B'B}) > 0 \rightarrow \text{right-oriented}$$

2.4. ABCD - tetrahedron



Consider the reference frames:

$$K_A = (A, \vec{AB}, \vec{AC}, \vec{AD}), \quad A = (\vec{AB}, \vec{AC}, \vec{AD})$$

$$K_B = (B, \vec{BA}, \vec{BC}, \vec{BD})$$

$$K'_A = (A, \vec{BA}, \vec{BD}, \vec{BC})$$

Define:

- Coords of A, B, C, D in all 3 K 's

- $M_{K_A K'_A}, M_{K_B K_A}$

- The orientation of the 3 bases w/ respect to each other

$$a) [ABCD]_{K_A} = \begin{pmatrix} \vec{AB} & \vec{AC} & \vec{AD} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$[ABCD]_{K_B} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$[ABCD]_{K_A'} = \begin{pmatrix} 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

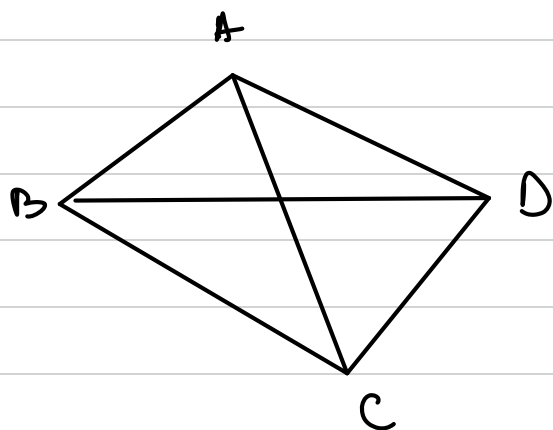
$$b) M_{K_A K_A'} = \begin{pmatrix} [BA]_{K_A} & [BD]_{K_A} & [BC]_{K_A} \\ -1 & -1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$\det M_{K_A K_A'} = 1 > 0 \Rightarrow K_A, K_A'$ have the same orientation

$$M_{K_B K_A} = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\det M_{K_B K_A} = -1 < 0 \Rightarrow K_A, K_B$ - opposite orientation

2.95. What is the # of contention frames one could construct w/ the vertices of a tetrahedron



$$(P, \vec{v}_1, \vec{v}_2, \vec{v}_3)$$

$$P \in \{A, B, C, D\}$$

$$\vec{v}_i \in \{\vec{xy} \mid x \neq y, x, y \in \{A, B, C, D\}\}$$

$$\vec{v}_{1,2,3} \text{ lin. indep.}$$

$$\begin{array}{ccc}
 4 & \cdot \left(4 \cdot 2 \cdot \underset{\substack{\text{6 sides}}}{C_6^2}} \right) & \cdot (2 \cdot 3) \\
 \downarrow & & \downarrow \\
 \text{\# of choices w/ the origin} & & \text{\# of choices for } \vec{v}_3 \\
 & \downarrow & \\
 & \text{\# of choices for } (\vec{v}_1, \vec{v}_2) &
 \end{array}$$

$$= 4 \cdot 8 \cdot 15 \cdot 6 = 2880$$

